SEC P vs NP log 2026-05-11

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-11 13:30 UTC

SEC P vs NP — daily log 2026-05-11

32 cycles recorded

Homotopy Group Rank Inverse Proportional to Disjointness Communication Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Topology (Homotopy Theory) × Disjointness Communication Complexity
- **Recorded**: 2026-05-11 00:01 UTC
- **Entry ID**: `c0f03be34ae3`

Statement

For any bipartite graph $G = (U \cup V, E)$ with $|U| = |V| = n$, the rank of the first homotopy group $\pi_1(\Gamma\_G)$ of its communication matrix Γ_G is $\Theta(1/CC(G))$, where CC(G) is the deterministic communication complexity of disjointness on G.

Rationale

Homotopy invariants capture global structural constraints of communication matrices, which may correlate with inherent complexity barriers. The fundamental group's rank reflects non-trivial loops in the matrix's topology, potentially mirroring the information-theoretic cost of coordination.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a1218782.py", line 86, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a1218782.py", line 68, in run_trial
    M = clique_incidence_matrix(G)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a1218782.py", line 37, in clique_incide
    M[u][v+n] = 1
    ~~~~^^^^^
```

```
IndexError: list assignment index out of range
```

Judge reasoning

Test crashed with IndexError, preventing data collection. Support condition cannot be evaluated without successful trials. | next: Fix the clique_incidence_matrix implementation to handle index bounds properly

---

Diophantine Approximation Error Inversely Proportional to Communication Discrepancy in XOR-Functions

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Diophantine Approximation × Communication Complexity of XOR-Functions
- **Recorded**: 2026-05-11 00:27 UTC
- **Entry ID**: `e98d1643eeaa`

Statement

For an XOR-function f: {0,1}^n × {0,1}^n → {0,1}, let D(f) denote its communication discrepancy and ε(f) denote the minimal approximation error of its Fourier coefficients by rational numbers with denominator ≤ n^2. Then D(f) ≥ Ω(1/ε(f) · log n).

Rationale

Diophantine approximation quantifies how well real numbers can be approximated by rationals, while discrepancy measures imbalance in communication matrices. Linking them could reveal structural constraints on XOR-functions' Fourier spectra, potentially separating complexity classes via approximation-theoretic barriers.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `1.91s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9b764fd6.py", line 101, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9b764fd6.py", line 67, in run_trial
    D_f = communication_discrepancy(xor_function, n)
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9b764fd6.py", line 52, in communication
    D += abs(f[(x, y)] - f[(x + 1) % (2**n)][(y + 1) % (2**n)])
                         ~^^^^^^^^^^^^^^^^^^
KeyError: 1
```

Judge reasoning

Test crashed with KeyError during computation of communication discrepancy, preventing validation of conjecture | next: Fix the KeyError in communication_discrepancy function by ensuring all (x,y) pairs are properly initialized in the XOR-function's domain

Hypergraph Matching Complexity for Monotone DNF

- **Verdict**: INCONCLUSIVE
- **Bridge**: Hypergraph Theory × Monotone DNF Formulas
- **Recorded**: 2026-05-11 00:37 UTC
- **Entry ID**: 5de245e2396b

Statement

For any monotone DNF formula with m terms over n variables, the maximum matching size μ satisfies: (i) $\mu \leq \log n + 2\sqrt{(m)}$ for $m \leq n^2$, (ii) $\mu \geq n/2$ for k-CLIQUE DNFs with $k \geq 2\sqrt{n}$.

Rationale

Hypergraph matching captures structural limitations of DNFs; its growth rate could expose inherent complexity gaps between polynomial-size DNFs and k-CLIQUE's exponential requirements. The conjecture links hypergraph expansion properties to circuit size lower bounds.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.11s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5734af53.py", line 94, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5734af53.py", line 68, in run_trial
    matching_size = len(greedy_matching(hypergraph))
                        ^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5734af53.py", line 50, in greedy_matchi
    matching.append(list(hypergraph_copy[max_edge_index]))
                         ~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^
TypeError: list indices must be integers or slices, not NoneType
```

Judge reasoning

Test crashed due to TypeError, preventing result collection. Pre-registered support condition cannot be evaluated without successful trials. | next: Debug the hypergraph indexing error in test_5734af53.py and re-run experiments

Additive Energy Threshold for ACC⁰ Circuit Size in CNF Formulas

- **Verdict**: INCONCLUSIVE
- **Bridge**: Additive Combinatorics × ACC⁰ Circuit Size for CNF Formulas
- **Recorded**: 2026-05-11 00:43 UTC
- **Entry ID**: 0553f6013ae8

Statement

For any CNF formula Φ with n variables, if the additive energy of the set of satisfying assignments S is $\Omega(n^3)$, then Φ cannot be computed by an ACC^0 circuit of size $o(n^2)$.

Rationale

Additive energy measures the structure of the solution set; ACC^0 circuits have limited capacity to capture such complex structures, leading to a lower bound on circuit size.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_89dca52f.py", line 83, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_89dca52f.py", line 63, in run_trial
    energy = add_energy(cnf, assignment)
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_89dca52f.py", line 49, in add_energy
    for x, y, z, w in combinations(range(n), 4):
                                         ^
NameError: name 'n' is not defined
```

Judge reasoning

Test crashed due to undefined variable 'n' in the add_energy function, preventing data collection. | next: Fix the NameError in test_89dca52f.py by defining 'n' before using it in combinations

---

SOS Refutation Size Bounded by Curve Genus of CSP Constraints

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Geometry (Curves over Finite Fields) × SOS Refutation Size for CSP Instances
- **Recorded**: 2026-05-11 00:50 UTC
- **Entry ID**: `66fe027cc0e4`

Statement

For a CSP instance over GF(2^n) with constraints defining a smooth projective curve C of genus g, the SOS refutation size is at most 2^{O(g log n)}. Conversely, if the curve has genus g ≥ log n, the SOS refutation size exceeds 2^{Ω(g)}.

Rationale

The genus captures the topological complexity of the solution space, influencing the number of moments required for SOS to capture contradictions. High-genus curves may encode inherently complex interactions between constraints, necessitating exponential refutation depth.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.03s`

```
9465729, 'instances_tested': 30, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'SOS Refutation Size', 'metric_value': 12.896247699465729, 'instances_tested'
TRIAL: {'metric_name': 'SOS Refutation Size', 'metric_value': 12.896247699465729, 'instances_tested'
TRIAL: {'metric_name': 'SOS Refutation Size', 'metric_value': 12.896247699465729, 'instances_tested'
TRIAL: {'metric_name': 'SOS Refutation Size', 'metric_value': 12.896247699465729, 'instances_tested'
TRIAL: {'metric_name': 'SOS Refutation Size', 'metric_value': 12.896247699465729, 'instances_tested'
TRIAL: {'metric_name': 'SOS Refutation Size', 'metric_value': 12.896247699465729, 'instances_tested'
TRIAL: {'metric_name': 'SOS Refutation Size', 'metric_value': 12.896247699465729, 'instances_tested'
TRIAL: {'metric_name': 'SOS Refutation Size', 'metric_value': 12.896247699465729, 'instances_tested'
TRIAL: {'metric_name': 'SOS Refutation Size', 'metric_value': 12.896247699465729, 'instances_tested'
TRIAL: {'metric_name': 'SOS Refutation Size', 'metric_value': 12.896247699465729, 'instances_tested'
TRIAL: {'metric_name': 'SOS Refutation Size', 'metric_value': 12.896247699465729, 'instances_tested'
TRIAL: {'metric_name': 'SOS Refutation Size', 'metric_value': 12.896247699465729, 'instances_tested'
RESULT: SUPPORTED mean=12.896247699465729 std=0.0 support_fraction=1.0
```

Judge reasoning

Safety rail: critic_challenge | original judge: All 30 tested instances satisfy the conjecture with consistent metric values and no counterexamples found. | next: Test larger genus values and higher n to validate the exponential bounds

---

Tutte Polynomial Evaluation Bounded by Monotone k-CLIQUE Circuit Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Tutte Polynomial Theory × Monotone Circuit Size for k-CLIQUE
- **Recorded**: 2026-05-11 01:05 UTC
- **Entry ID**: `954a5c8e8ffd`

Statement

For any k-CLIQUE instance on n vertices, the Tutte polynomial evaluated at (2, 1) satisfies T(2,1) ≥ Ω(n^{k/2}) and T(1,2) ≥ Ω(n^{k/2}), where T(x,y) is the Tutte polynomial of the graphic matroid induced by the complete graph K_n.

Rationale

The Tutte polynomial encodes matroid structure, and its evaluation at (2,1) and (1,2) captures connectivity properties. For k-CLIQUE, these values may correlate with the exponential growth of monotone circuit size due to the combinatorial explosion in clique enumeration.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_12ec44ec.py", line 99, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_12ec44ec.py", line 72, in run_trial
    T21, T12 = get_tutte_values(graph, n // 2 + 1)
               ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_12ec44ec.py", line 61, in get_tutte_val
    T21 = tutte_polynomial(graph)
          ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_12ec44ec.py", line 29, in tutte_polynom
    if (graph,) in memo:
       ^^^^^^^^^^^^^^^^
TypeError: unhashable type: 'collections.defaultdict'
```

Judge reasoning

Test crashed due to unhashable defaultdict graph input | next: Fix graph representation to use hashable structures like frozenset for memoization

---

Young Tableau Count Exponential Gap in Disjointness Communication Matrices

- **Verdict**: INCONCLUSIVE
- **Bridge**: Algebraic Combinatorics (Standard Young Tableau Count) × Communication Complexity (Disjointness Problem)
- **Recorded**: 2026-05-11 01:20 UTC
- **Entry ID**: 2b7d955e959e

Statement

For a random disjointness instance with n variables, the number of standard Young tableaux of shape λ (where λ is the partition derived from the communication matrix) is exponentially larger than for the corresponding determinant matrix, implying a lower bound on the communication complexity.

Rationale

The exponential gap in tableau counts reflects the combinatorial structure of the communication matrix, which could be leveraged to derive lower bounds via representation-theoretic methods.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_525950f5.py", line 86, in <module>
```

```
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_525950f5.py", line 66, in run_trial
    det_tableau_count = count_standard_young_tableaux(det_partition)
                        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_525950f5.py", line 58, in count_standar
    return hook_length_formula(partition)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_525950f5.py", line 26, in hook_length_f
    for cell, size in enumerate(row):
                      ^^^^^^^^^^^^^^
TypeError: 'int' object is not iterable
```

Judge reasoning

Test crashed due to TypeError, preventing data collection. The crash indicates a code bug rather than a conjecture failure. | next: Fix the hook_length_formula implementation to handle partitions correctly

---

Fourier Coefficient Sum Inverse Proportional to Disjointness Discrepancy

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Fourier Analysis of Boolean Functions × Communication Complexity of DISJOINTNESS
- **Recorded**: 2026-05-11 01:40 UTC
- **Entry ID**: `982d20a21495`

Statement

For every n-variable DISJOINTNESS instance, the sum of absolute Fourier coefficients of its characteristic function satisfies $\Sigma|\hat{f}(S)| = \Theta(1/\Delta)$, where Δ is the discrepancy of the communication matrix. Equality holds when the instance is a balanced complete bipartite graph.

Rationale

Fourier coefficients capture correlation with parity functions, while discrepancy measures communication complexity. Their inverse proportionality suggests hidden structural alignment between spectral properties and combinatorial communication lower bounds.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.06s`


```
ed": 463, "metric_name": "Fourier Coefficient Sum", "metric_value": 466.0, "instances_tested": 40, "c
TRIAL: {"seed": 503, "metric_name": "Fourier Coefficient Sum", "metric_value": 548.0, "instances_test
TRIAL: {"seed": 547, "metric_name": "Fourier Coefficient Sum", "metric_value": 438.0, "instances_test
TRIAL: {"seed": 593, "metric_name": "Fourier Coefficient Sum", "metric_value": 476.0, "instances_test
TRIAL: {"seed": 631, "metric_name": "Fourier Coefficient Sum", "metric_value": 454.0, "instances_test
TRIAL: {"seed": 677, "metric_name": "Fourier Coefficient Sum", "metric_value": 504.0, "instances_test
TRIAL: {"seed": 727, "metric_name": "Fourier Coefficient Sum", "metric_value": 564.0, "instances_test
TRIAL: {"seed": 773, "metric_name": "Fourier Coefficient Sum", "metric_value": 440.0, "instances_test
```

```
TRIAL: {"seed": 821, "metric_name": "Fourier Coefficient Sum", "metric_value": 632.0, "instances_test
TRIAL: {"seed": 877, "metric_name": "Fourier Coefficient Sum", "metric_value": 504.0, "instances_test
TRIAL: {"seed": 929, "metric_name": "Fourier Coefficient Sum", "metric_value": 616.0, "instances_test
RESULT: FALSIFIED counterexample="mapping_undefined" first_failing_seed=11
```

Judge reasoning

Safety rail: critic_challenge_falsified | original: Test results show conjecture_holds is False for multiple seeds with counterexample 'mapping_undefined'. The presence of any counterexample meets the f | next: Investigate the 'mapping_undefined' counterexample to determine specific DISJOINTNESS instances violating the conjecture

---

SOS Refutation Degree Bounded by Clause Density for Random 3-SAT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Real Algebraic Geometry × 3-SAT Instances
- **Recorded**: 2026-05-11 01:51 UTC
- **Entry ID**: `52c162cccd9b`

Statement

For a random 3-SAT instance with n variables and m clauses, the minimal SOS degree required to refute it is $\Theta(\sqrt{m})$.

Rationale

The SOS hierarchy's refutation degree captures the algebraic complexity of constraint systems. By linking it to clause density, we expose geometric structure in SAT solving that could yield new lower bounds for refutation algorithms.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
569495584, got inf"}
TRIAL: {"seed": 593, "metric_name": "SOS Degree", "metric_value": inf, "instances_tested": 1, "conjec
TRIAL: {"seed": 631, "metric_name": "SOS Degree", "metric_value": inf, "instances_tested": 1, "conjec
TRIAL: {"seed": 677, "metric_name": "SOS Degree", "metric_value": inf, "instances_tested": 1, "conjec
TRIAL: {"seed": 727, "metric_name": "SOS Degree", "metric_value": inf, "instances_tested": 1, "conjec
TRIAL: {"seed": 773, "metric_name": "SOS Degree", "metric_value": inf, "instances_tested": 1, "conjec
TRIAL: {"seed": 821, "metric_name": "SOS Degree", "metric_value": inf, "instances_tested": 1, "conjec
TRIAL: {"seed": 877, "metric_name": "SOS Degree", "metric_value": inf, "instances_tested": 1, "conjec
TRIAL: {"seed": 929, "metric_name": "SOS Degree", "metric_value": inf, "instances_tested": 1, "conjec

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0e7c05f5.py", line 93, in <module>
    first_failing_seed = next(r['seed'] for r in results if not r['conjecture_holds'])
                         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0e7c05f5.py", line 93, in <genexpr>
    first_failing_seed = next(r['seed'] for r in results if not r['conjecture_holds']
```

Judge reasoning

parse_fail | next: NONE

Schatten p-Norm Lower Bound on Disjointness Communication Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Noncommutative L^p Geometry × Communication Complexity of DISJOINTNESS
- **Recorded**: 2026-05-11 01:59 UTC
- **Entry ID**: `aa045d952985`

Statement

For any n ≥ 1, the Schatten p-norm of the communication matrix for the DISJOINTNESS problem on n-bit inputs is Ω(n), where p is the dual exponent satisfying 1/p + 1/q = 1 for the optimal lower bound.

Rationale

The Schatten p-norm captures the operator structure of the communication matrix, which is tightly linked to the required information-theoretic resources in multi-party communication. For DISJOINTNESS, the norm's growth with n reflects the inherent difficulty of coordinating disjointness checks.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.14s`


```
nstances_tested': 30, 'conjecture_holds': False, 'counterexample': 'Schatten p-
norm does not scale linearly with n'}
TRIAL: {'metric_name': 'Schatten p-Norm', 'metric_value': 10.011753727979853, 'instances_tested': 30
norm does not scale linearly with n'}
TRIAL: {'metric_name': 'Schatten p-Norm', 'metric_value': 9.731746236597688, 'instances_tested': 30,
norm does not scale linearly with n'}
TRIAL: {'metric_name': 'Schatten p-Norm', 'metric_value': 10.011658675891052, 'instances_tested': 30
norm does not scale linearly with n'}
TRIAL: {'metric_name': 'Schatten p-Norm', 'metric_value': 9.672856354946585, 'instances_tested': 30,
norm does not scale linearly with n'}
TRIAL: {'metric_name': 'Schatten p-Norm', 'metric_value': 8.734210406193428, 'instances_tested': 30,
norm does not scale linearly with n'}
TRIAL: {'metric_name': 'Schatten p-Norm', 'metric_value': 9.586306087041859, 'instances_tested': 30,
norm does not scale linearly with n'}
TRIAL: {'metric_name': 'Schatten p-Norm', 'metric_value': 9.752608783128863, 'instances_tested': 30,
norm does not scale linearly with n'}
TRIAL: {'metric_name': 'Schatten p-Norm', 'metric_value': 9.97251009486534, 'instances_tested': 30,
norm does not scale linearly with n'}
TRIAL: {'metric_name': 'Schatten p-Norm', 'metric_value': 9.914781880120312, 'instances_tested': 30,
norm does not scale linearly with n'}

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_be69cc65.p
```

Judge reasoning

Test crashed with ZeroDivisionError, preventing reliable results. Pre-registered support condition cannot be evaluated without valid data. | next: Fix the test to handle empty results and re-run with sufficient sample size

---

Matroid Rank Lower Bound for Monotone k-CLIQUE

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Matroid Theory × Monotone Circuit Size for k-CLIQUE
- **Recorded**: 2026-05-11 03:08 UTC
- **Entry ID**: `cf5188cf0224`

Statement

For a monotone DNF formula φ, let M_φ be the matroid whose ground set is clauses of φ and rank function r(S) = max{|S′| : S′ ⊆ S, S′ is a collection of pairwise disjoint clauses}. Then r(M_φ) = O(log n) for all φ with circuit size ≤ n^c, but r(M_φ) = Ω(n) for all k-CLIQUE instances with k ≥ 3.

Rationale

Matroid rank captures combinatorial independence, which may mirror the structural constraints of monotone circuits. High rank for k-CLIQUE suggests inherent complexity, while low rank for general formulas hints at decomposability, offering a novel angle to circuit lower bounds.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.02s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Increase timeout duration and re-run test with extended computation limits

---

Symmetric Power Decomposition Complexity Gap for 3-SAT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Schur-Weyl Duality × 3-SAT Instances
- **Recorded**: 2026-05-11 03:39 UTC
- **Entry ID**: `9f467f497a7b`

Statement

For a 3-SAT instance with n variables, the number of irreducible components in the symmetric power decomposition of its clause-variable incidence matrix is Ω(2^n) larger than for the corresponding determinant’s decomposition.

Rationale

Schur-Weyl duality decomposes tensor powers into irreducible representations; permanent's symmetric nature yields exponentially more components than determinant's alternating structure, exposing algebraic barriers to polynomial identity testing.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.39s`

```
7942604329759159257642539931170845074868151383943896496785650941364228155170896708389178684652312
TRIAL: {'metric_name': 'Symmetric Power Decomposition Complexity Gap', 'metric_value': 0, 'instances_
```

Judge reasoning

Test crashed before producing valid data, rendering the metric value unreliable. | next: Re-run test with increased time limits and verify code stability

---

Specht Module Decomposition Exponential Gap Between Permanent and Determinant Polynomials

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic combinatorics (Specht modules, symmetric functions) × SOS refutation degree for polynomial CSP instances
- **Recorded**: 2026-05-11 04:26 UTC
- **Entry ID**: `5afbe3121f1c`

Statement

For random n-variable polynomial CSP instances derived from the permanent and determinant, the SOS refutation degree of the permanent instance is exponentially larger than that of the determinant instance for all $n \geq 2$.

Rationale

The exponential growth in the number of Specht modules in the permanent's decomposition suggests higher structural complexity, which could correlate with increased SOS refutation requirements.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.05s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Run test with extended timeout and higher resource allocation to complete execution

---

Persistent Homology Gap in Read-Twice BPs for IP_2

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Persistent Homology (Topological Data Analysis) × Read-Twice Branching Programs
- **Recorded**: 2026-05-11 04:32 UTC
- **Entry ID**: `f6f1e06ee5ff`

Statement

For read-twice BP P computing IP_2, the maximum persistence of 1-dimensional homology classes in the simplicial complex built from P's state transition graph satisfies persistence(P) ≥ Ω(n). For general read-twice BPs computing any function f, persistence(P) ≤ O(log size(P))

Rationale

Persistent homology captures topological robustness of state transition structures. IP_2's inherent symmetry creates persistent holes in its BP structure, while general functions' transitions collapse faster. This links algebraic topology to BP size complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.04s`


```
dom': False, 'counterexample': 'IP_2 BP failed: persistence(P) < Ω(n)\nRandom function BP failed: pers
TRIAL: {'metric_name': 'persistence', 'metric_value_ip2': 3, 'metric_value_random': 3, 'instances_te
TRIAL: {'metric_name': 'persistence', 'metric_value_ip2': 3, 'metric_value_random': 3, 'instances_te
TRIAL: {'metric_name': 'persistence', 'metric_value_ip2': 4, 'metric_value_random': 4, 'instances_te
TRIAL: {'metric_name': 'persistence', 'metric_value_ip2': 4, 'metric_value_random': 4, 'instances_te
TRIAL: {'metric_name': 'persistence', 'metric_value_ip2': 9, 'metric_value_random': 9, 'instances_te
TRIAL: {'metric_name': 'persistence', 'metric_value_ip2': 9, 'metric_value_random': 9, 'instances_te
TRIAL: {'metric_name': 'persistence', 'metric_value_ip2': 19, 'metric_value_random': 19, 'instances_
```


Judge reasoning

Insufficient instances_tested=2 to validate asymptotic bounds. Counterexamples may stem from small-n behavior rather than general complexity. | next: Test with larger n (e.g., n≥1000) to observe asymptotic scaling of persistence metrics

---

Free Entropy Gap in Read-Twice BP Transition Matrices

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Free Probability Theory × Read-Twice Branching Programs
- **Recorded**: 2026-05-11 04:50 UTC
- **Entry ID**: `9858410e3069`

Statement

Let P be a read-twice BP computing a function f: {0,1}^n → {0,1}. Define M_P as the n×n matrix where M_P[i][j] = (number of paths from root to leaf i that read bit j twice). The free entropy φ(M_P) satisfies φ(M_P) ≤ log(size(P)) + O(√n), but for IP_2 (inner product mod 2), φ(M_P) ≥ Ω(n log n).

Rationale

Free entropy quantifies non-commutative randomness in matrix laws; read-twice BPs exhibit structured non-commutativity via repeated bit access. This gap could reveal inherent complexity in repeated access patterns, analogous to how Shannon entropy bounds communication complexity.

Novelty

- Judge: `INCONCLUSIVE` over 5 arXiv hits

Top hits consulted: - [1201.0716v2] Concavification of free entropy - [1510.05486v2] Two types of branching programs with bounded repetition that cannot efficiently compute monotone 3-CNFs - [1902.03873v2] Analogues of Entropy in Bi-Free Probability Theory: Non-Microstate - [1809.06500v3] Range entropy: A bridge between signal complexity and self-similarity - [1902.03874v3] Analogues of Entropy in Bi-Free Probability Theory: Microstates

Empirical Test

- exit code: `1`, elapsed: `1.25s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5342c042.py", line 90, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5342c042.py", line 64, in run_trial
    R = r_transform(M_P)
        ^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5342c042.py", line 44, in r_transform
    if M[i][j] > 0:
       ~~~~^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed with IndexError before producing data; cannot assess support fraction or counterexamples | next: Debug r_transform implementation to handle matrix indexing properly

---

Fundamental Group Rank Inverse Proportional to Resolution Proof Size for Tseitin Formulas

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Topology (Fundamental Groups of Graphs) × Resolution Proof Size for Tseitin Formulas
- **Recorded**: 2026-05-11 05:05 UTC
- **Entry ID**: `2905aa033f8a`

Statement

For a Tseitin formula derived from a connected expander graph with m edges and n vertices, the resolution proof size is at least 2^{c * (m - n + 1)} for some absolute constant c > 0. The exponent equals the rank of the fundamental group π_1 of the graph.

Rationale

Expander graphs' high connectivity implies nontrivial π_1 structure, which may encode combinatorial obstructions to resolution proofs. The exponential dependency suggests that topological complexity directly limits proof efficiency, offering a novel algebraic-topological barrier to resolution.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
RIAL: {"seed": 593, "metric_name": "resolution_proof_size", "metric_value": 0, "instances_tested": 1
TRIAL: {"seed": 631, "metric_name": "resolution_proof_size", "metric_value": 0, "instances_tested": 
TRIAL: {"seed": 677, "metric_name": "resolution_proof_size", "metric_value": 0, "instances_tested": 
TRIAL: {"seed": 727, "metric_name": "resolution_proof_size", "metric_value": 0, "instances_tested": 
TRIAL: {"seed": 773, "metric_name": "resolution_proof_size", "metric_value": 0, "instances_tested": 
TRIAL: {"seed": 821, "metric_name": "resolution_proof_size", "metric_value": 0, "instances_tested": 
TRIAL: {"seed": 877, "metric_name": "resolution_proof_size", "metric_value": 0, "instances_tested": 
TRIAL: {"seed": 929, "metric_name": "resolution_proof_size", "metric_value": 0, "instances_tested": 

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_32d3e164.py", line 134, in <module>
    first_failing_seed = next((r['seed'] for r in results if not r['conjecture_holds']), None)
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_32d3e164.py", line 134, in <genexpr>
    first_failing_seed = next((r['seed'] for r in results if not r['conjecture_holds']), None)
                                            ~^^^^^^^^
KeyError: 'seed'
```

Judge reasoning

Test crashed before completing required trials; insufficient data to evaluate support fraction or counterexamples | next: Fix the test code's KeyError handling and re-run with complete seed coverage

---

Real Rank of Moment Matrix for Random Max-CUT Instances is Linear in n

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Real Algebraic Geometry (Moment Matrices, Positivstellensatz) × SOS Degree for Max-CUT
- **Recorded**: 2026-05-11 05:26 UTC
- **Entry ID**: `d71faf6d27d8`

Statement

For a random Max-CUT instance with n variables and m clauses, the real rank of its moment matrix is Θ(n), implying that the minimal SOS degree required to approximate the maximum cut is Θ(n).

Rationale

The real rank of the moment matrix captures the complexity of the polynomial system defining the Max-CUT instance. A linear real rank suggests that the SOS hierarchy requires linear degree to capture the solution, which could relate to the inherent structure of the problem.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_96b1ab55.py", line 109, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_96b1ab55.py", line 82, in run_trial
    rank = matrix_rank(M)
           ^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_96b1ab55.py", line 37, in matrix_rank
    A = gaussian_elimination(A)
        ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_96b1ab55.py", line 30, in gaussian_elim
    factor = A[j][i] / A[i][i]
             ~~~~~~~~^~~~~~~~~
ZeroDivisionError: float division by zero
```

Judge reasoning

Test crashed with ZeroDivisionError, preventing evaluation of rank properties | next: Fix numerical stability in Gaussian elimination or test with different parameter ranges

---

Finite-Field Rank Threshold for ACC⁰ Circuit Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Linear Algebra over Finite Fields × ACC⁰ Circuit Size
- **Recorded**: 2026-05-11 05:56 UTC
- **Entry ID**: `de7db3666529`

Statement

Let f: {0,1}^n → {0,1} be a Boolean function represented by a GF(2)-matrix M of rank r. Then, if r > n^c for some c ∈ (0,1), f requires ACC⁰ circuits of size Ω(n^{1/c})

Rationale

Finite-field rank captures algebraic structure that may inherently resist ACC^0 computation. The conjecture links matrix rank to circuit size via a power-law threshold, offering a new algebraic barrier for explicit functions.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8796f7bd.py", line 91, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8796f7bd.py", line 53, in run_trial
    r = rank(matrix)
        ^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8796f7bd.py", line 39, in rank
    reduced_matrix = gaussian_elimination(matrix)
                     ^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8796f7bd.py", line 30, in gaussian_elim
    matrix[i][j] /= factor
   File "/usr/lib/python3.12/fractions.py", line 615, in forward
    return monomorphic_operator(a, b)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^
   File "/usr/lib/python3.12/fractions.py", line 763, in _div
    raise ZeroDivisionError('Fraction(%s, 0)' % db)
ZeroDivisionError: Fraction(1, 0)
```

Judge reasoning

Test crashed during Gaussian elimination due to ZeroDivisionError, preventing data collection | next: Fix GF(2) arithmetic implementation to use bitwise operations instead of Fraction class

---

Secant Rank Lower Bound for Disjointness Tensors

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Geometry (Secant Varieties of Veronese Reembeddings) × Communication Complexity of DISJOINTNESS
- **Recorded**: 2026-05-11 06:02 UTC
- **Entry ID**: `3b212abe5c22`

Statement

For a DISJOINTNESS matrix M ∈ {0,1}^{n×n}, let T_M ∈ R^{n×n×2} be the tensor formed by stacking M and its transpose. The secant rank of T_M is Ω(n).

Rationale

Secant rank captures the minimal number of rank-1 tensors needed to span T_M. High secant rank implies geometric complexity, which may correlate with communication complexity via the tensor product structure of DISJOINTNESS.

Novelty

- Judge: `NOVEL` over 3 arXiv hits

Top hits consulted: - [1011.5867v2] Secant Varieties of Segre–Veronese Varieties - [1012.3563v4] Secant varieties to high degree Veronese reembeddings, catalecticant matrices and smoothable Gorenstein schemes - [1503.01099v2] Singularities of secant varieties

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c67fe947.py", line 81, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c67fe947.py", line 66, in run_trial
    secant_rank_value = secant_rank(T_M)
                        ^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c67fe947.py", line 53, in secant_rank
    U, V = rank_1_decomposition(T)
           ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c67fe947.py", line 44, in rank_1_decomp
    U_new = matrix_multiply(U, T)
            ^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c67fe947.py", line 37, in matrix_multip
    C[i][j] += A[i][l] * B[l][j]
               ~~~~~~~~^~~~~~~~~
TypeError: can't multiply sequence by non-int of type 'float'
```

Judge reasoning

Test crashed due to a type error in matrix multiplication, preventing any data collection. The pre-registered support condition cannot be evaluated without successful trials. | next: Fix the matrix_multiply function's type handling and re-run tests with multiple seeds to collect empirical evidence.

Real Dimension of Parity-Constraint Variety Bounds AC^0 Depth

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Real Algebraic Geometry × AC^0 Circuits Computing PARITY
- **Recorded**: 2026-05-11 06:31 UTC
- **Entry ID**: `5db94ffcbe67`

Statement

For any AC^0 circuit C computing PARITY on n inputs, the real dimension of the semialgebraic set defined by C's constraints is $\Omega(\log \text{size}(C))$.

Rationale

PARITY's linear structure forces its constraint variety to have low real dimension, but AC^0 circuits with deeper structure may encode higher-dimensional varieties. This conjecture links algebraic geometry's real dimension to circuit complexity via a concrete mapping of gates to polynomial inequalities.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.27s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_833e0728.py", line 91, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_833e0728.py", line 61, in run_trial
    circuit = generate_ac0_circuit(n)
              ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_833e0728.py", line 27, in generate_ac0_
    inputs = [random.randint(0, i-1) for _ in range(random.randint(2, 3))]
              ^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/random.py", line 336, in randint
    return self.randrange(a, b+1)
           ^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/random.py", line 319, in randrange
    raise ValueError(f"empty range in randrange({start}, {stop})")
ValueError: empty range in randrange(0, 0)
```

Judge reasoning

Test crashed due to invalid randrange parameters, preventing data collection | next: Fix the generate_ac0_circuit function to ensure valid random range parameters

---

Cheeger Constant Inverse Proportional to Extended Frege Proof Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Spectral Geometry × Extended Frege Proof Size
- **Recorded**: 2026-05-11 07:16 UTC
- **Entry ID**: `bd4677253c0f`

Statement

For a 3-CNF instance Φ with n variables, the Cheeger constant h(Φ) of its hypergraph representation satisfies h(Φ) = Θ(1 / size(Π)), where Π is the minimal Extended Frege proof of Φ.

Rationale

The Cheeger constant, a spectral geometry invariant, captures the expansion properties of the hypergraph. In proof complexity, expansion properties might correlate with the required proof size, as highly connected structures may necessitate longer proofs.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5952b1b6.py", line 103, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5952b1b6.py", line 86, in run_trial
    h_phi = cheeger_constant(adj_list)
            ^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5952b1b6.py", line 52, in cheeger_const
    laplacian = [[d - A[i][j] for j, d in enumerate(degree)] + [0] * (n -
len(adj_list[i])) for i, adj_list in enumerate(adj_list)]
                      ^
NameError: name 'A' is not defined
```

Judge reasoning

Test crashed due to undefined variable 'A' in cheeger_constant calculation, preventing data collection | next: Fix the NameError in the cheeger_constant function by properly defining adjacency matrix 'A' before computing Laplacian

---

Communication Matrix Rank Lower Bounds ABP Size for Boolean Functions

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Matrix Rank over Finite Fields × ABP Size
- **Recorded**: 2026-05-11 07:42 UTC
- **Entry ID**: `0071c0003c24`

Statement

For any Boolean function f: {0,1}^n × {0,1}^n → {0,1}, the size of the minimal ABP computing f is at least the rank of its communication matrix over GF(2).

Rationale

The rank of the communication matrix captures linear dependencies in the function's behavior, which may impose structural constraints on the ABP's required size. ABPs inherently model computations as polynomial sequences, and low-rank matrices might correspond to more structured polynomials, potentially requiring larger ABPs to capture the necessary complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.33s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out without producing results, preventing evaluation of support fraction or counterexamples. | next: Run test with extended timeout and debug logging to identify crash cause

---

Invariant Generator Count Bounds SOS Refutation Degree for Symmetric CSPs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Invariant Theory × SOS Refutation Degree for CSPs
- **Recorded**: 2026-05-11 08:00 UTC
- **Entry ID**: `6972f4aaf7c4`

Statement

For a CSP instance with symmetry group G, the SOS refutation degree is at most the number of invariant generators of its constraint polynomials under G. Formally, if P is a polynomial ideal defining the CSP's feasible solutions, then deg_SOS(P) ≤ |InvGen(P, G)|.

Rationale

Symmetries in CSPs often reduce the effective complexity by collapsing orbits of variables. Invariant theory provides tools to decompose constraint polynomials into symmetry-invariant components, which may directly limit the SOS hierarchy's required degree via orbit-stabilizer principles.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1abb6691.py", line 117, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1abb6691.py", line 94, in run_trial
    if is_invariant(poly, G):
       ^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1abb6691.py", line 60, in is_invariant
    if poly[i] != poly[g[0][i]]:
       ~~~~^^^
TypeError: 'int' object is not subscriptable
```

Judge reasoning

Test crashed due to TypeError in subscripting integer object, preventing data collection | next: Fix the is_invariant function to handle polynomial representations correctly

---

Newton Polytope Volume Inverse Proportional to SOS Degree for Max-CUT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Geometry (Newton Polytopes) × SOS Degree for Max-CUT Instances
- **Recorded**: 2026-05-11 08:32 UTC
- **Entry ID**: `9dba9c55f385`

Statement

For a Max-CUT instance with n variables, the volume of the Newton polytope of its corresponding degree-d SOS relaxation polynomial is Θ(1/d^2). The volume is inversely proportional to d^2, with the constant depending on the instance's edge density.

Rationale

Newton polytopes encode the monomial structure of polynomials, which directly reflects the constraint geometry of SOS relaxations. The volume captures the 'spread' of monomials, which may limit the efficiency of higher-degree relaxations by increasing the number of terms in the Gram matrix.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.04s`


```
Volume', 'metric_value': 0, 'instances_tested': 9, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'Volume', 'metric_value': 0, 'instances_tested': 9, 'conjecture_holds': True,
TRIAL: {'metric_name': 'Volume', 'metric_value': 0, 'instances_tested': 9, 'conjecture_holds': True,
TRIAL: {'metric_name': 'Volume', 'metric_value': 0, 'instances_tested': 9, 'conjecture_holds': True,
TRIAL: {'metric_name': 'Volume', 'metric_value': 0, 'instances_tested': 9, 'conjecture_holds': True,
TRIAL: {'metric_name': 'Volume', 'metric_value': 0, 'instances_tested': 9, 'conjecture_holds': True,
TRIAL: {'metric_name': 'Volume', 'metric_value': 0, 'instances_tested': 9, 'conjecture_holds': True,
TRIAL: {'metric_name': 'Volume', 'metric_value': 0, 'instances_tested': 9, 'conjecture_holds': True,
TRIAL: {'metric_name': 'Volume', 'metric_value': 0, 'instances_tested': 9, 'conjecture_holds': True,
TRIAL: {'metric_name': 'Volume', 'metric_value': 0, 'instances_tested': 9, 'conjecture_holds': True,
TRIAL: {'metric_name': 'Volume', 'metric_value': 0, 'instances_tested': 9, 'conjecture_holds': True,
TRIAL: {'metric_name': 'Volume', 'metric_value': 0, 'instances_tested': 9, 'conjecture_holds': True,
TRIAL: {'metric_name': 'Volume', 'metric_value': 0, 'instances_tested': 9, 'conjecture_holds': True,
TRIAL: {'metric_name': 'Volume', 'metric_value': 0, 'instances_tested': 9, 'conjecture_holds': True,
TRIAL: {'metric_name': 'Volume', 'metric_value': 0, 'instances_tested': 9, 'conjecture_holds': True,
TRIAL: {'metric_name': 'Volume', 'metric_value': 0, 'instances_tested': 9, 'conjecture_holds': True,
RESULT: SUPPORTED mean=0.0 std=0.0 support_fraction=1.0
```


Judge reasoning

Tested instances show perfect support but suffer from n ≤ 15 (trivial scaling), metric saturation at 0, and potential selection bias. Pre-registered criteria require ≥80% seeds, which is met, but critic challenges invalidate confidence. | next: Test with larger n (≥30) and adversarial edge density configurations to probe non-trivial volume scaling behavior

---

Krull Dimension Lower Bounds SOS Refutation Degree for 3-SAT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Geometry (Krull Dimension of Ideals) × SOS Refutation Degree for 3-SAT Instances
- **Recorded**: 2026-05-11 08:45 UTC
- **Entry ID**: `e6b8ed98a856`

Statement

For a 3-SAT instance with m clauses over n variables, the SOS refutation degree required to prove unsatisfiability is at least the Krull dimension of the ideal generated by the clause polynomials (each clause is mapped to a polynomial via (1 - x_i)(1 - x_j)(1 - x_k) = 0 for a clause (x_i ∨ x_j ∨ x_k)).

Rationale

The Krull dimension captures the 'geometric complexity' of the solution space's algebraic variety. SOS refutations must account for this structure, suggesting a lower bound tied to the variety's intrinsic dimension. This bridges algebraic geometry with proof complexity via polynomial optimization.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_097a0593.py", line 124, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_097a0593.py", line 103, in run_trial
    m = random.randint(n, min(2 * n, 30))
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/random.py", line 336, in randint
    return self.randrange(a, b+1)
           ^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/random.py", line 319, in randrange
    raise ValueError(f"empty range in randrange({start}, {stop})")
ValueError: empty range in randrange(33, 31)
```

Judge reasoning

Test crashed due to invalid randrange parameters (33, 31), preventing data collection. Support condition cannot be evaluated without valid results. | next: Fix the test parameters to ensure valid randrange ranges (stop >= start) and re-run with proper seed handling

---

Finite-Field Rank Threshold for ACC^0 Circuit Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Finite Field Geometry × ACC^0 Circuit Size for Sipser Function
- **Recorded**: 2026-05-11 10:35 UTC
- **Entry ID**: `ac07c338962c`

Statement

For the Sipser function on n variables, the rank of its truth-table matrix over GF(2) is $\Omega(2^{n/2})$, and this rank is exponentially larger than the minimal ACC^0 circuit size computing it.

Rationale

The high rank of the truth-table matrix over GF(2) reflects the function's algebraic structure, which may inherently resist efficient ACC^0 computation due to the limited depth of such circuits.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.13s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out without producing results; insufficient data to evaluate support fraction or detect counterexamples. | next: Increase timeout duration and re-run experiment with deterministic seed configuration

---

Gonality Lower Bounds for Disjointness Communication Matrices

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Curves over Finite Fields × Communication Complexity of DISJOINTNESS
- **Recorded**: 2026-05-11 11:01 UTC
- **Entry ID**: `5e12fc0dd31f`

Statement

For a DISJOINTNESS matrix M over GF(2), associate it with a curve C over GF(2^k) via its incidence structure. The gonality $\gamma(C)$ satisfies $\gamma(C) \geq \Omega(\sqrt{n})$ if M requires $\Omega(\sqrt{n})$ communication complexity.

Rationale

Gonality measures the minimal degree of a morphism from a curve to the projective line, capturing structural complexity. High gonality may mirror the combinatorial rigidity of disjointness matrices, linking algebraic geometry to communication lower bounds.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.10s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing validation of support fraction or counterexample detection. | next: Increase timeout duration and re-run with stricter resource limits to ensure completion

---

Projective Plane Disjointness Complexity Bounded by Line Count

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Finite Geometry (Projective Planes) × Communication Complexity (Set Disjointness)
- **Recorded**: 2026-05-11 11:36 UTC
- **Entry ID**: `8b9459572783`

Statement

For a projective plane of order q, the communication complexity of the set disjointness problem where each player receives a subset of lines and must determine if their sets are disjoint is $\Theta(q^2 \log q)$. This is because the number of lines is $\Theta(q^2)$, and the problem's structure requires checking for intersections, which is inherently tied to the geometry's parameters.

Rationale

The projective plane's structure enforces specific intersection properties between lines, which can be leveraged to derive lower bounds on communication complexity. The geometry's parameters directly influence the required information exchange between players.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_08a91518.py", line 96, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_08a91518.py", line 60, in run_trial
    subset1 = set(random.sample(lines, random.randint(1, num_lines // 2)))
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
TypeError: unhashable type: 'set'
```

Judge reasoning

Test crashed due to unhashable type error, preventing data collection. The conjecture's validity cannot be assessed without successful execution. | next: Fix the TypeError in test code by using frozensets or tuples instead of sets for hashability

---

Noncommutative Rank Bounded by Disjointness Communication Complexity

- **Verdict**: `INCONCLUSIVE`

- **Bridge**: Noncommutative Geometry × Communication Complexity of Disjointness
- **Recorded**: 2026-05-11 12:25 UTC
- **Entry ID**: `8ac0ba31a882`

Statement

For any n×n Boolean matrix M arising from the disjointness problem, its noncommutative rank (minimal number of rank-1 matrices needed to express M as a sum in the matrix algebra over the ring of 2×2 matrices) satisfies rank_noncommutative(M) ≤ 2^{n/2} / (disjointness_commsize(M))

Rationale

Noncommutative rank captures algebraic structure beyond classical linear algebra, potentially exposing hidden symmetries in communication matrices. Disjointness has optimal Ω(n) lower bounds, suggesting a inverse relationship between algebraic complexity and communication cost.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `2.55s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_73f8d0b0.py", line 71, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_73f8d0b0.py", line 54, in run_trial
    rank_noncommutative = noncommutative_rank(M)
                          ^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_73f8d0b0.py", line 44, in noncommutativ
    A = [[A_combination[i][j] for j in range(k)] for i in range(k)]
          ~~~~~~~~~~~~~~~~^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed with IndexError before producing results, preventing evaluation of conjecture validity | next: Fix the noncommutative_rank function to handle matrix dimensions properly and re-run tests

---

Kronecker Coefficient Asymmetry in Permanent vs Determinant Decompositions

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Schur-Weyl Duality × Homogeneous Polynomial Complexity
- **Recorded**: 2026-05-11 12:48 UTC
- **Entry ID**: `cb66523ac3ef`

Statement

For all n ≥ 2, the number of distinct irreducible representations in the decomposition of perm_n under the symmetric group S_n exceeds the number for det_m where m < n^{1.5}, and this count is preserved under linear substitutions.

Rationale

Schur-Weyl duality decomposes tensor powers into symmetric group irreps. Permanent's decomposition has more distinct components than determinant's due to its non-GL(n)-invariant structure, which could mirror complexity gaps via representation-theoretic invariants.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a65f3c06.py", line 80, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a65f3c06.py", line 57, in run_trial
    perm_count = count_irreducible_components([n], n)
                 ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_a65f3c06.py", line 47, in count_irreduc
    for part in decomposition:
TypeError: 'int' object is not iterable
```

Judge reasoning

Test crashed due to TypeError before collecting data; support condition cannot be evaluated | next: Fix decomposition handling in count_irreducible_components to handle integer vs iterable cases

---

Plethysm Coefficient Growth and Permanent Circuit Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Plethysm of Symmetric Functions × Boolean Circuit Size for Permanent Polynomial
- **Recorded**: 2026-05-11 13:20 UTC
- **Entry ID**: `743e042ea636`

Statement

For each n ≥ 1, the plethysm coefficient of s_{(n)} ∘ s_{(n)} is exponentially larger than the minimal circuit size of the n×n permanent polynomial over GF(2).

Rationale

The plethysm coefficient captures the multiplicity of irreducible representations in the decomposition, which may reflect the inherent complexity of expressing the permanent as a combination of simpler polynomials. If this coefficient grows exponentially, it suggests that the permanent requires exponentially large circuits, aligning with known lower bounds.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 0.03s


```
257874440.0, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'plethysm_coefficie
TRIAL: {'metric_name': 'plethysm_coefficient', 'metric_value': 9075135300.0, 'instances_tested': 1,
TRIAL: {'metric_name': 'plethysm_coefficient', 'metric_value': 155117520.0, 'instances_tested': 1, '
TRIAL: {'metric_name': 'plethysm_coefficient', 'metric_value': 3432.0, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'plethysm_coefficient', 'metric_value': 126410606437752.0, 'instances_tested
TRIAL: {'metric_name': 'plethysm_coefficient', 'metric_value': 8233430727600.0, 'instances_tested':
TRIAL: {'metric_name': 'plethysm_coefficient', 'metric_value': 48620.0, 'instances_tested': 1, 'conj
TRIAL: {'metric_name': 'plethysm_coefficient', 'metric_value': 6.892620648693261e+21, 'instances_tes
TRIAL: {'metric_name': 'plethysm_coefficient', 'metric_value': 601080390.0, 'instances_tested': 1, '
TRIAL: {'metric_name': 'plethysm_coefficient', 'metric_value': 4.425125402768368e+20, 'instances_tes
TRIAL: {'metric_name': 'plethysm_coefficient', 'metric_value': 35345263800.0, 'instances_tested': 1,
TRIAL: {'metric_name': 'plethysm_coefficient', 'metric_value': 2333606220.0, 'instances_tested': 1,
RESULT: FALSIFIED counterexample="plethysm_coefficient_too_large" first_failing_seed=11
```


Judge reasoning

parse_fail | next: NONE

Real Radical Rank Lower Bounds SOS Degree for Max-CUT

- **Verdict**: INCONCLUSIVE
- **Bridge**: Real Algebraic Geometry (Real Radical Ideals) × SOS Degree for Max-CUT Instances
- **Recorded**: 2026-05-11 13:30 UTC
- **Entry ID**: e37540873b70

Statement

For any Max-CUT instance with n variables, the SOS degree required to achieve Ω(1) approximation is at least the real radical rank of the ideal defining the instance. Formally, if I is the ideal generated by the quadratic forms of the graph, then deg_SOS(I) ≥ rank_real_radical(I).

Rationale

Real radical ideals capture the algebraic structure of real solutions to polynomial systems. By linking their rank to SOS degree, we expose a geometric constraint on the complexity of semidefinite relaxations, potentially revealing inherent limitations in approximation algorithms.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 19.10s


```
re_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'sos_degree', 'metric_value': 0, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'sos_degree', 'metric_value': 0, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'sos_degree', 'metric_value': 0, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'sos_degree', 'metric_value': 0, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'sos_degree', 'metric_value': 0, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'sos_degree', 'metric_value': 0, 'instances_tested': 1, 'conjecture_holds': T
```

```
TRIAL: {'metric_name': 'sos_degree', 'metric_value': 0, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'sos_degree', 'metric_value': 0, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'sos_degree', 'metric_value': 0, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'sos_degree', 'metric_value': 0, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'sos_degree', 'metric_value': 0, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'sos_degree', 'metric_value': 0, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'sos_degree', 'metric_value': 0, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'sos_degree', 'metric_value': 0, 'instances_tested': 1, 'conjecture_holds': T
TRIAL: {'metric_name': 'sos_degree', 'metric_value': 0, 'instances_tested': 1, 'conjecture_holds': T
RESULT: SUPPORTED mean=0.0 std=0.0 support_fraction=1.0
```

Judge reasoning

The test shows 100% support but uses trivial instances ($n \leq 1$) where SOS degree is zero, making the bound vacuous. The critic correctly challenges the lack of adversarial cases with non-zero radical rank. | next: Test with larger n and non-trivial radical rank examples to validate the conjecture's generality